

Recentered Planar Joint Admission: A Scoped Search and Full-Spectrum Audit

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Abstract—Recentring a constrained endpoint search at a previously certified regular second-moment root yields 309 accepted steps in 424 attempts for a six-qubit planar interface. Sampled operator residual improves slightly, while the conservative unit-norm deadline increases. Exact rational contraction certifies the selected root, but full-spectrum enclosures prove complete half-diamond error greater than $1/16$ at every deadline. Recentring establishes neither operational admission nor resource advantage. The target error 2^{-16} remains open.

Index Terms—Autonomous quantum control, joint admission, two-local Hamiltonian, spectral certificate.

I. CONTRACT AND DISCLOSED EXPLORATION

Registers are (A, O, u, v, w, z) ; four modes represent $(00, 01, 10, 11)$. The interface is

$$E|ab\rangle = |ab\rangle_{AO}|1_{00}\rangle_M, \quad (1)$$

$$W|ab\rangle = |00\rangle_{AO}|1_{ab}\rangle_M. \quad (2)$$

All four columns require coherent admission with one common phase and arbitrary spectators. Single excitation is an endpoint condition, not an invariant transit-sector claim. Full physical dimension is 64. The planar graph permits ten pairs $\{Au, Aw, Az, Ou, Ov, Ow, Oz, uw, uv, wz\}$ with maximum degree four. There are 108 nonidentity Hermitian Pauli components of support at most two, including fields; real coefficients allow complex entries. The model adds no measurement, reset, bath or scheduled pulses. Launch and observation remain external assumptions; thermodynamic refresh is unproved.

An exact first-moment kernel gives $G(\theta) = \sum_k (B\theta)_k P_k$ with 97 real coordinates. Fifteen independent quadratic equations match second compressed moments. The parent [1] certifies a regular algebraic root, a local 82-dimensional branch and a dyadic center. We recenter at that numerical center using the parent's fifteen dependent ports. The center is not substituted for the root in an exact claim.

The protocol is frozen before this run, after exposure to the parent's outcomes and chart boundary. The enlarged free-coordinate box and budget are disclosed tuning, not independent prediction. Physical graph and norm budget remain fixed. This is exploratory engineering; no independent confirmatory endpoint dataset exists.

MQM development note v 0.1, October 6, 2026. ORCID: 0009-0002-5813-0724. Exploratory continuation; independent review and novelty comparison pending. Physical promotion: 0.

II. REDUCED SEARCH AND IMPLEMENTATION CHECKS

Split $\theta = (y, z)$ into 15 dependent and 82 free coordinates. Newton retraction solves $q(y, z) = 0$ numerically; the tangent lift obeys

$$\frac{dy}{dz} = -J_y^{-1} J_z. \quad (3)$$

The objective retains all coherent columns:

$$R = e^{-iG(\theta)} E - e^{i\phi} W, \quad f = \frac{1}{2} \|R\|_F^2. \quad (4)$$

An 83-column reduced Jacobian combines 82 tangent directions and one common phase. Damped Gauss–Newton proposes steps; numerical retraction and guards decide acceptance. Frobenius cost is an optimization surrogate, not the certified complete-channel error.

The recentered free box has infinity radius four, versus one previously; the attempt cap is 1000. Other guards remain: Jacobian condition at most 10^7 , 20 Newton iterations, dependent step at most one, normalized moment tolerance $2 \cdot 10^{-15}$ and Newton-step tolerance $2 \cdot 10^{-13}$ after at least two iterations. Extended precision evaluates polynomial residuals. These checks do not certify a continuous exact path or globally regular chart.

With $L = \sum_k |c_k|$, implement $H = G/L$ at $T = L$, giving full norm at most one on all 64 dimensions. The conservative deadline cap is 512; no optimal centered norm is claimed. At two probes, 166 central-difference reduced derivative columns use step 10^{-6} . Maximum discrepancy is $9.44\text{e-}08$, with tangent defect $4.54\text{e-}15$. These are numerical implementation checks, not independent review.

III. OUTCOME AND EXACT ALL-TIME EXCLUSION

Seed 2026100606 yields 424 attempts: 309 accepted and 115 rejected, comprising 79 failures to decrease cost and 36 free-box violations. The damping cap terminates before 1000 attempts. Neither stationarity nor global optimality follows. The best sampled operator-residual point is selected among accepted points, while acceptance minimizes Frobenius cost. The residual decreases by approximately $3.57 \cdot 10^{-5}$ while the conservative deadline grows. This establishes no resource saving, measured device performance or comparative advantage over other clocks or timed controls.

Freeze the selected numerical coordinates as binary rationals and refine a 15-coordinate slice. Exact rational bounds for $\mathcal{B}(y) = y - R_0 q(y)$ certify

$$\beta < 1, \quad \eta + \beta r < r, \quad r = 2^{-48}. \quad (5)$$

TABLE I
MATCHED GRAPH AND NORM-BUDGET COMPARISON

Quantity	Parent	Recentered
Physical qubits	6	6
Allowed pairs	10	10
Attempts	158	424
Accepted steps	100	309
Sampled operator residual	0.222881819	0.222846101
Conservative deadline	358.894545	373.851497
All-time half-diamond floor	$> 1/16$	$> 1/16$

Banach contraction gives a unique real algebraic root in this fixed slice box. The remaining 82 coordinates are exact dyadics. The selected minor remains nonsingular, giving a local 82-dimensional zero manifold. First and second moments match exactly; full norm at most one and a deadline below 512 are certified. These facts do not imply admission.

Exact Gaussian-integer spectral residuals enclose the dyadic center's true projectors. The root-box norm debit transfers the enclosure through separated spectral gaps. A rational upper enclosure of

$$S = \sum_{\lambda} |\text{Tr}(W^{\dagger} \Pi_{\lambda} E)| \quad (6)$$

gives, using a maximally entangled logical/reference input,

$$\frac{1}{2} \|\mathcal{U}_t - \mathcal{W}\|_{\diamond} \geq \sqrt{1 - (S/4)^2} > 1/16. \quad (7)$$

The enclosed radical diagnostic is 0.103574. The bound holds at every deadline, including scalar shifts and uniform norm scaling. It excludes this root only, not the planar family.

IV. SCREENING, REPRODUCIBILITY AND NEXT GATE

The parent's shared-return criterion requires Krylov dimension at most 61 for perfect admission. For simple spectrum and complete error at most $\epsilon < 1$, the three smallest shared-launch weights must sum to at most $(\epsilon/(1-\epsilon))^2$. At target 2^{-16} the threshold is exactly $1/65535^2$. The floating center sum passes this necessary screen but is not an enclosed root-weight certificate; the all-time floor still rejects admission.

The release contains 252 scoped exact checks and 166 numerical derivative checks, protocol, code, dependencies, manifests and parent hashes. The exact tally includes rerun inherited geometry and task-data checks; it is not 252 new theorems. Automated checks establish neither independent review nor empirical evidence.

Repeated recentering in this basin gives marginal improvement. The next exploratory gate should compare independent regular charts and monitor spectral overlap alongside endpoint residual under the same budget. Selected success requires exact feasibility and complete error at most 2^{-16} on a nonzero readout window. Native admission remains OPEN; encoded SWAP-core composition is gated. Physical promotion remains 0. Code correction, fault tolerance, scalability, quantum advantage and autonomous thermal refresh are unestablished.

REFERENCES

- [1] B. W. Mayes, "Retracted Admission Search and a Shared-Return Spectral Criterion," MQM development note v 0.1, Oct.6,2026, immutable replay release; independent review pending.